

Cold-leaching extraction. A new methodology for obtaining inhibitory substances produced by bacteria in solid media.

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Abstract

Inhibitory substances can be obtained and purified by different methodologies. Free cell supernatants obtained by centrifugation and filtration are commonly used for evaluating antimicrobial metabolites produced by bacteria. In this work, a new methodology for obtaining antimicrobial metabolites is proposed, especially implemented for bacteria which are not able to produce these compounds in liquid media; based on the use of nitrocellulose filter membranes and solid culture media.

Culture medium for extraction

Extraction can be performed by using any solid media. It would be ideal the use of minimum culture media, so it does not interfere in the purification steps. However, it is important to test the

producer strain in different conditions as the medium composition, pH, temperature, and cell mass could interfere in the metabolite production [1–3]. In this methodology, the nutritionally rich Luria-Bertani medium was used, which contains per liter of medium: peptone 10 g, yeast extract 0.5 g, and sodium chloride 10 g [4].

Cold leaching extraction

The cold-leaching extraction is performed by using the agar where the producer strain was grown previously. For this, 100 μL of an overnight culture with around 6.5×10^7 UFC/mL of the producer strain are placed on sterile nitrocellulose filter membranes with 0.22 μm pore size [5]; previously placed on the surface of small agar plates with approximately 10 mL of culture media. Once the nitrocellulose membrane surface is dry, plates are incubated for 30 h at 30 °C.

After incubation, membranes are removed, and all the agar is cut with a sterile scalpel and deposited into a conical bottom centrifuge tube previously filled with the solvent to evaluate; with proportion of 3 g of agar sample per milliliter of solvent. Absolute ethanol previously filtered was used for this purpose.

After recollection, tubes are incubated for 48 h at 4 °C in order to achieve extraction. Subsequently, tubes are centrifuged for 12 minutes at 4 °C and 5000 rpm to separate the supernatant in new tubes. Moreover, extracts can be evaporated using a rotary evaporator to collect a more concentrated sample. If this last step is performed, we strongly recommend corroborating the antimicrobial activity in the aqueous and solvent phases, to ensure that no substances of interest are lost in the solvent removed.

It is worth mentioning that solvents of different polarity can be used in this assay when the chemical nature of the substances to be extracted is unknown.

Once the samples are finally collected, they are sterilized with 0.22 μm pore size filters, and evaluated by the double-layer agar assay [6] or the agar well diffusion method [7] to determine their inhibitory capacity (Fig 1). In addition, arbitrary units (AU/mL) can be determined [8]. Control samples can be obtained by extracting from agar where there was no bacterial growth.

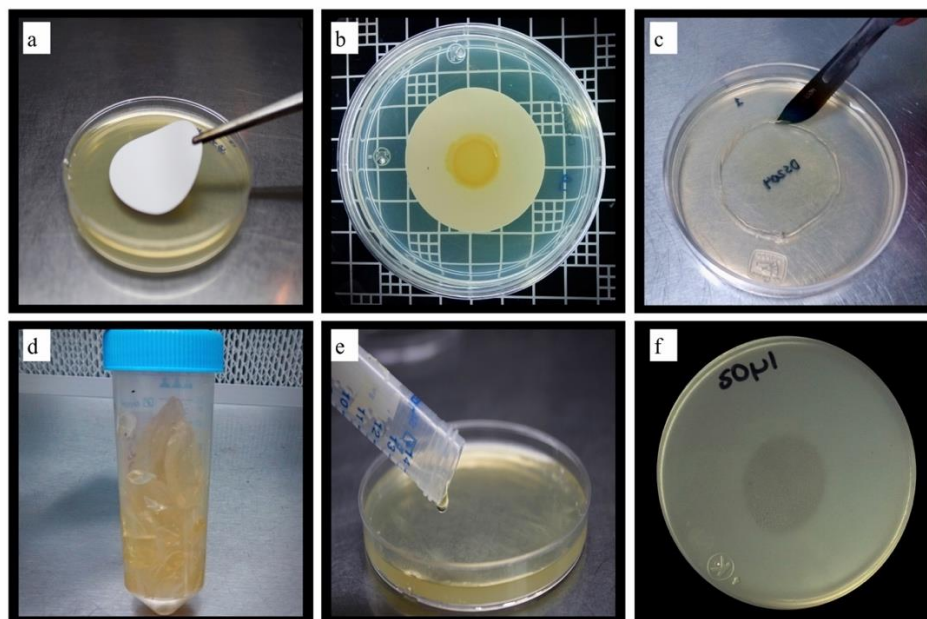


Fig 1. Cold leaching extraction method. a) Nitrocellulose filter membranes placed on the agar surface; b) Bacterial growth after incubation period; c) agar cut, and d) transfer to cold ethanol for incubation at 4 °C; e) antimicrobial extract assay; f) inhibition halo formed after performing the double-layer agar methodology.

Remarks

The methodology described here was developed for a bacterial strain that showed no antimicrobial activity in liquid media and could be used as a new opportunity to discover inhibitory substances in microorganisms that have not shown antimicrobial activity in traditional processes.

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